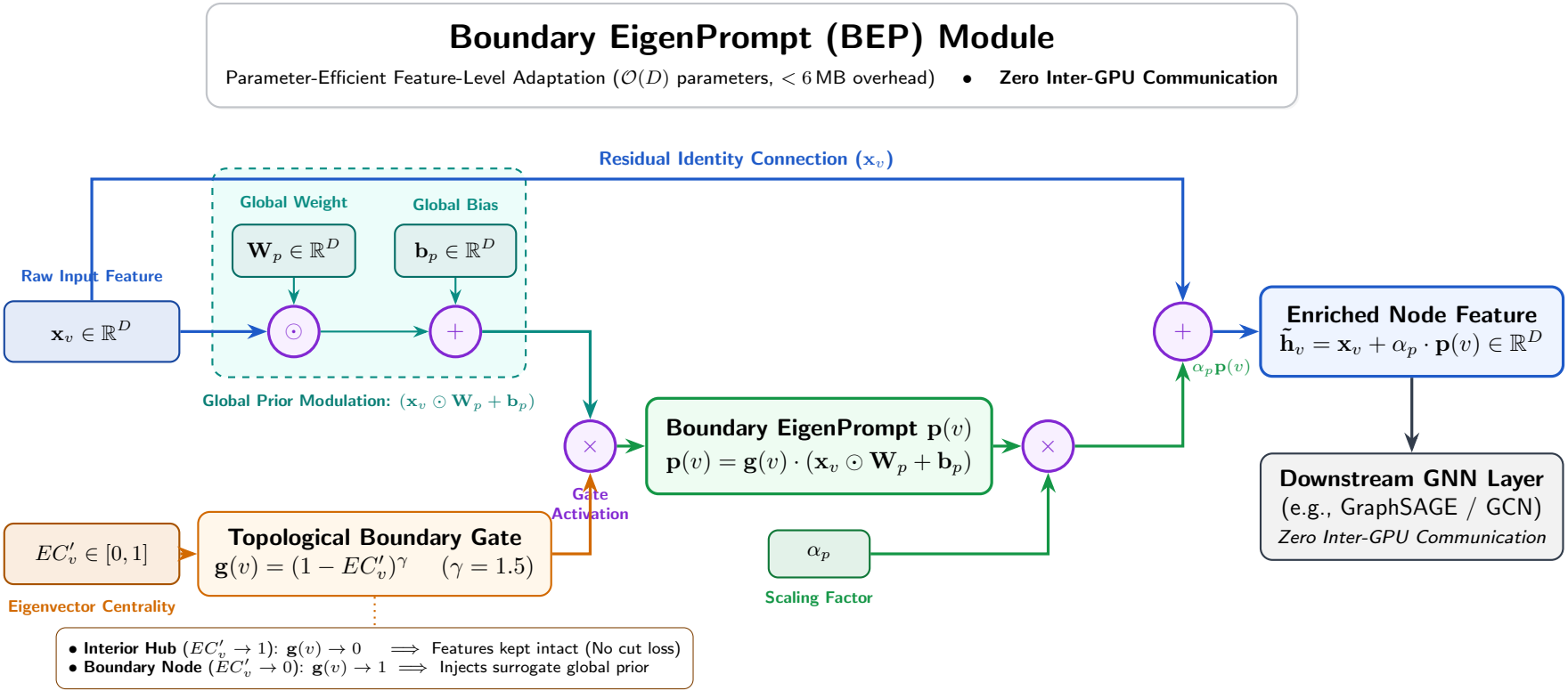

Boundary EigenPrompt (BEP) Module
Parameter-Efficient Feature-Level Adaptation ($\mathcal{O}(D)$ parameters, $< 6$ MB overhead) • Zero Inter-GPU Communication
Residual Identity Connection ($\mathbf{x}_v$)
Global Weight
Global Bias
$\mathbf{W}_p \in \mathbb{R}^D$
$\mathbf{b}_p \in \mathbb{R}^D$
Raw Input Feature
$\mathbf{x}_v \in \mathbb{R}^D$
Global Prior Modulation: $(\mathbf{x}_v \odot \mathbf{W}_p + \mathbf{b}_p)$
Gate Activation
Boundary EigenPrompt $\mathbf{p}(v)$
$\mathbf{p}(v) = \mathbf{g}(v) \cdot (\mathbf{x}_v \odot \mathbf{W}_p + \mathbf{b}_p)$
$\alpha_p \mathbf{p}(v)$
Enriched Node Feature
$\tilde{\mathbf{h}}_v = \mathbf{x}_v + \alpha_p \cdot \mathbf{p}(v) \in \mathbb{R}^D$
Downstream GNN Layer
(e.g., GraphSAGE / GCN)
Zero Inter-GPU Communication
$EC'_v \in [0,1]$
Eigenvector Centrality
Topological Boundary Gate
$\mathbf{g}(v) = (1 - EC'_v)^\gamma \quad (\gamma = 1.5)$
$\alpha_p$
Scaling Factor
• Interior Hub ($EC'_v \to 1$): $\mathbf{g}(v) \to 0 \implies$ Features kept intact (No cut loss)
• Boundary Node ($EC'_v \to 0$): $\mathbf{g}(v) \to 1 \implies$ Injects surrogate global prior